

First-year Analysis Examination
Part Two
August 2021

Answer FOUR questions in detail.
State carefully any results used without proof.

1. Let f_A be the indicator function of the countably infinite set $A \subseteq [0, 1]$: that is, $f_A(t)$ is 1 when $t \in A$ and 0 when $t \in [0, 1] \setminus A$.
 - (i) Exhibit (with brief justification) such an A for which f_A is Riemann integrable, or prove that no such A exists.
 - (ii) Exhibit (with brief justification) such an A for which f_A is not Riemann integrable, or prove that no such A exists.
2. For each $n > 0$ let the function $f_n : X \rightarrow \mathbb{R}$ be continuous at all but finitely many points. Prove that if $f_n \rightarrow f$ uniformly on X , then f is continuous at all but countably many points.
3. $(f_n : n > 0)$ is an equicontinuous sequence of real-valued functions on a compact space. Prove that if $f_n \rightarrow f$ pointwise then $f_n \rightarrow f$ uniformly.
4. Let $(f_n : n > 0)$ be a sequence of measurable functions on Ω . Prove that each of the following sets is measurable:
 - (i) $\{\omega \in \Omega : \text{the sequence } f_n(\omega) \text{ is eventually constant}\}$;
 - (ii) $\{\omega \in \Omega : \text{the values } f_n(\omega) \text{ are all different}\}$.
5. Let $(f_n : n > 0)$ be a sequence of non-negative integrable functions that converges pointwise to f . Prove that if

$$\int_{\Omega} f_n \, d\mu \rightarrow \int_{\Omega} f \, d\mu$$

then

$$\int_{\Omega} |f - f_n| \, d\mu \rightarrow 0.$$

[It might help to consider the positive part $(f - f_n)^+ = \max\{0, f - f_n\}$.]
